Practical No. 8

Aim: Write a java program demonstrating exception handelling.

Code:

Program 1:

```
public class ExceptionDemo {
    public static void main(String[] args) {
        int numerator = 10;
        int denominator = 0; // Division by zero triggers an ArithmeticException

        try {
            System.out.println("Attempting division...");
            int result = numerator / denominator; // Code that may throw an exception
            System.out.println("Result: " + result); // Skipped if an exception occurs
        }
        catch (ArithmeticException e) {
            // Handles division by zero
            System.out.println("Error: Cannot divide by zero!");
        }
        finally {
            // Always executes, regardless of whether an exception occurred
            System.out.println("Execution completed.");
        }
    }
}
```

Program 2:

```
import java.io.BufferedReader;
import java.io.FileReader;
import java.io.FileNotFoundException;
import java.io.IOException;

public class ReadFileDemo {
    public static void main(String[] args) {
        String filePath = "sample.txt"; // File path to read

        BufferedReader reader = null;

        try {
            // Attempt to open the file
            reader = new BufferedReader(new FileReader(filePath));

            String line;
            System.out.println("Reading file contents:");

            // Read line by line until reaching the end of the file (null)
            while ((line = reader.readLine()) != null) {
                System.out.println(line);
            }
```

```
        } catch (FileNotFoundException e) {
            // Specific catch for when the file does not exist at the given path
            System.out.println("Error: The specified file was not found -> " + filePath);

        } catch (IOException e) {
            // General catch for I/O errors (e.g., read failure, permission denied)
            System.out.println("Error: An I/O error occurred while reading the file.");
            e.printStackTrace();

        } finally {
            // Ensure the reader stream is closed to release system resources
            try {
                if (reader != null) {
                    reader.close();
                    System.out.println("\nFile resource closed successfully.");
                }
            } catch (IOException e) {
                System.out.println("Error closing the reader stream.");
            }
        }
    }
}
```

Output:

```
Attempting division...
ERROR!
Error: Cannot divide by zero!
Execution completed.
```

```
ERROR!
Error: The specified file was not found -> sample.txt
```